It is certifiably golden

A small sub-set of EDA tools, which stands a class apart, are classified as golden. Also known as sign-off quality, these tools are highly accurate and ease the end-to-end process from RTL to GDS in case of ASIC development and timing closure for field-programmable gate arrays. Several semiconductor foundries provide pre-evaluated reference methodologies, also called RM flows, consisting of suitable tools, versions and scripts in order to automate the entire design process.

Ruchir Dixit, technical director, Mentor Graphics, explains some of these golden features as, “We have a more accurate implementation, and we check for those implementations.” The tools provide an indication for problems, which can then be handled. With added features, support for how to fix those problems and how quickly can you fix it can be taken care of. He adds, “The system also supports systems with multiple PCBs and provide a larger design environment.”

have requirements for precision at about 0.1mm. The latest EDA tools support this requirement in order to bring about the best in design.

Digital electronics designs also require very high precision. Current solutions are prepared to support the best in design. “Composite EM technology in ADS platform maintains the same level of accuracy, and we have increased the performance by a large margin,” says Deepak R.K., general manager, EEs of EDA, Keysight Technologies India Pvt Ltd.

Special cases of board design. Wearables and other special use cases have increased the demand for flexible PCBs. “Rigid-flex is a technology for designers working on flexible PCBs,” says Tamanna. With this new feature, designers can verify parameters that might interfere with different movements and shapes of the product.

Some parameters to be kept in mind while designing flexible PCBs would be to not compromise on the physical and electrical integrity of the circuit. “Beyond certain parameters set while designing flexible PCBs, the board loses its integrity,” explains Dixit. Open source tools are also adding such features since wearables often employ flexible PCBs.

Working in groups. Dixit says, “Designs have become complex. Calibre provides provisions for breaking the complex board into individual components.” These individual components can then be used by different people. Today, tools offer live collaboration for multiple people working in real time.

Open source tools are also updating to the latest features with the group working feature being one. “The master-slave mode is for teams working on different levels of the project,” says Deshpande. This can also be used to monitor the progress of the design.

Improving the finer points. Making the interface attractive and easy to use certainly helps, and the industry